

Spectrophotometric Methods

Midterm Exam

Part I: Multiple Choice Questions (MCQs)

1. Which of the following terms describes the distance between two consecutive crests of a wave?
 - a) Frequency
 - b) Wave number
 - c) Wavelength
 - d) Energy
2. The relationship between wavelength (λ) and frequency (ν) is:
 - a) Directly proportional
 - b) Inversely proportional
 - c) Constant
 - d) No relationship
3. A Deuterium lamp is used as a light source for which region?
 - a) Visible
 - b) Infrared
 - c) Ultraviolet (UV)
 - d) Radio waves
4. According to Beer's Law, absorbance is directly proportional to:
 - a) Path length
 - b) Type of cuvette
 - c) Concentration
 - d) Transmittance
5. In a Spectrophotometer, the function of the Monochromator is to:
 - a) Convert light into an electrical signal
 - b) Select a specific wavelength from the light source
 - c) Hold the liquid sample
 - d) Increase light intensity
6. Why is the cuvette placed at a 90-degree angle in a Spectrofluorometer?
 - a) To reduce cost
 - b) To avoid measuring the incident light directly
 - c) To increase analysis speed
 - d) To allow IR passage
7. Which technique measures light emitted by thermally excited atoms?



- a) UV-Vis Spectrophotometry
 - b) Atomic Absorption Spectroscopy (AAS)
 - c) Atomic Emission Spectroscopy (AES)
 - d) IR Spectrophotometry
8. A solution containing all reagents except the analyte is called a:
- a) Standard solution
 - b) Test solution
 - c) Blank solution
 - d) Saturated solution
9. The process of converting a liquid sample into free atoms in the gas phase in AAS is:
- a) Absorption
 - b) Atomization
 - c) Excitation
 - d) Transmission
10. A major advantage of Atomic Absorption Spectroscopy (AAS) is its:
- a) Slow speed
 - b) Ability to measure only organic matter
 - c) High sensitivity for trace metal analysis
 - d) Inaccuracy in mineral testing

Part II: True or False

- 1. The energy of electromagnetic radiation is directly proportional to its frequency. ()
- 2. The visible light range is below 380 nm. ()
- 3. High Transmittance (T) indicates a very concentrated and dark solution. ()
- 4. Lambert's Law states that absorbance is proportional to the path length of light. ()
- 5. Quartz cuvettes are used for measurements in the UV region. ()
- 6. An absorption spectrum shows colored lines on a black background. ()
- 7. Emission Spectroscopy (AES) can be used to determine lead in petrol. ()
- 8. A Spectrofluorometer is generally more sensitive than a standard Spectrophotometer. ()
- 9. λ Max is the wavelength where a compound shows minimum absorbance. ()

Part III: Short Answer & Calculations

- 1. Mention two clinical applications of spectrophotometric methods.



2. What is the main difference in the light source between a Spectrophotometer and AAS?
3. Why is a 'Blank' solution essential before any measurement?
4. Briefly describe what happens to an electron during 'Atomic Absorption'.
5. List two disadvantages of Emission Spectroscopy.
6. Which elements are ideally measured using Flame Photometry?
7. A molecule excites at 450 nm and emits at 520 nm. Name the phenomenon and why the wavelength increased.
8. Calculate the concentration: Standard (10 mg/dL) Absorbance = 0.400, Test Absorbance = 0.200.
9. Compare Absorption and Emission spectra in terms of appearance.
10. List three components common to all spectrophotometric instruments.



Answer Key

Part I: MCQs

1. c | 2. b | 3. c | 4. c | 5. b | 6. b | 7. c | 8. c | 9. b | 10. c

Part II: True/False

1. T | 2. F | 3. F | 4. T | 5. T | 6. F | 7. T | 8. T | 9. F

Part III: Short Answers

1. Measuring glucose, cholesterol, enzymes, etc.
2. Spectrophotometer uses lamps (Tungsten/Deuterium); AAS uses Hollow Cathode Lamps (HCL).
3. To zero the instrument and account for absorbance by reagents/solvents.
4. It absorbs energy and moves from ground state to an excited state.
5. Expensive equipment, complicated procedures.
6. Alkali metals (Na, K, Li) and Alkaline earth metals (Ca, Mg).
7. Fluorescence; wavelength increased due to Stokes Shift (energy loss as heat).
8. $\text{Conc} = (0.200 / 0.400) * 10 = 5 \text{ mg/dL}$.
9. Absorption: Dark lines on colored background. Emission: Colored lines on dark background.
10. Light source, Monochromator, Detector.

